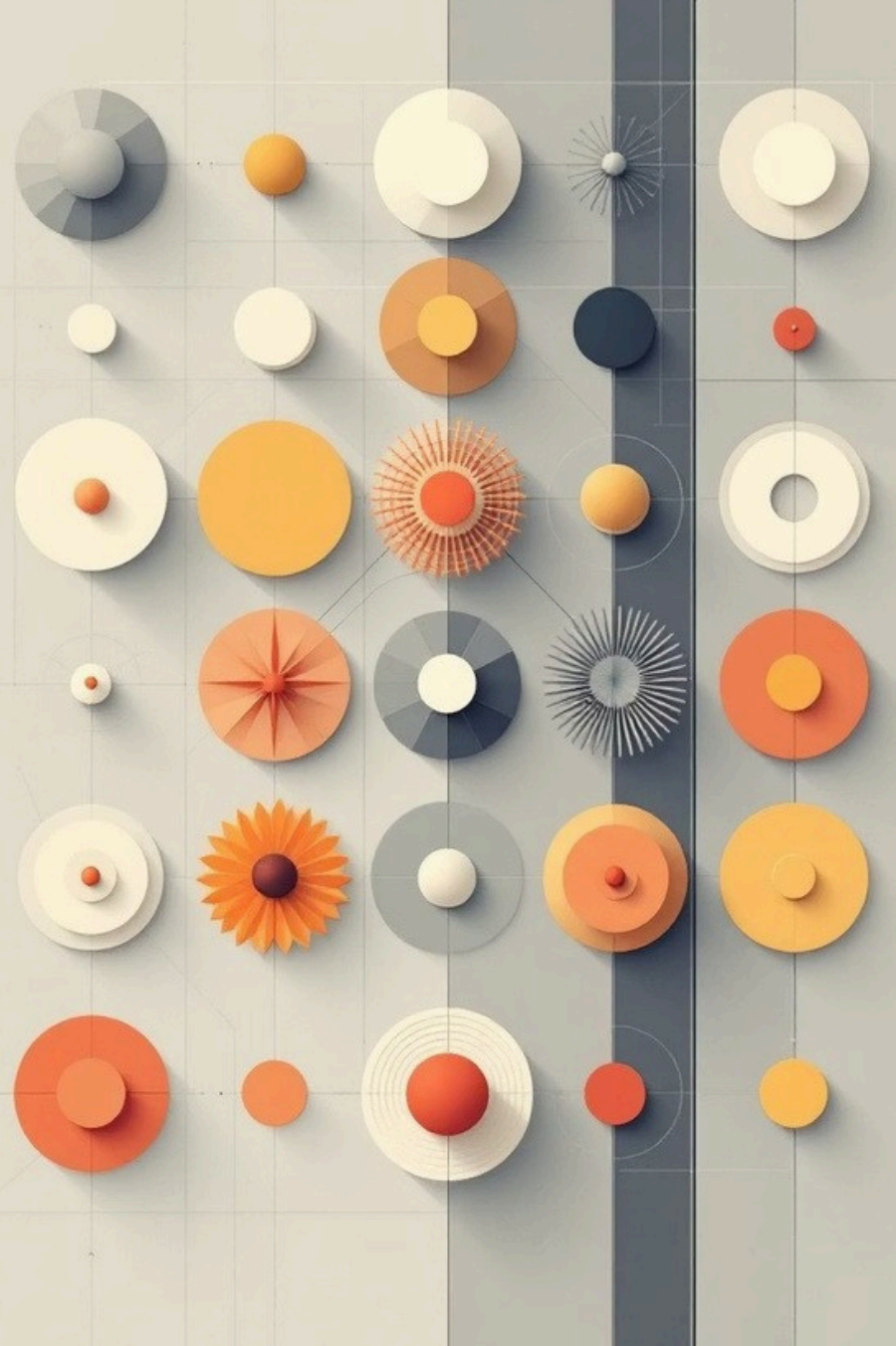




Constructing a Square Equivalent in Area to a Given Circle

A computational exploration of the Baudhayana Sulba Sutra method

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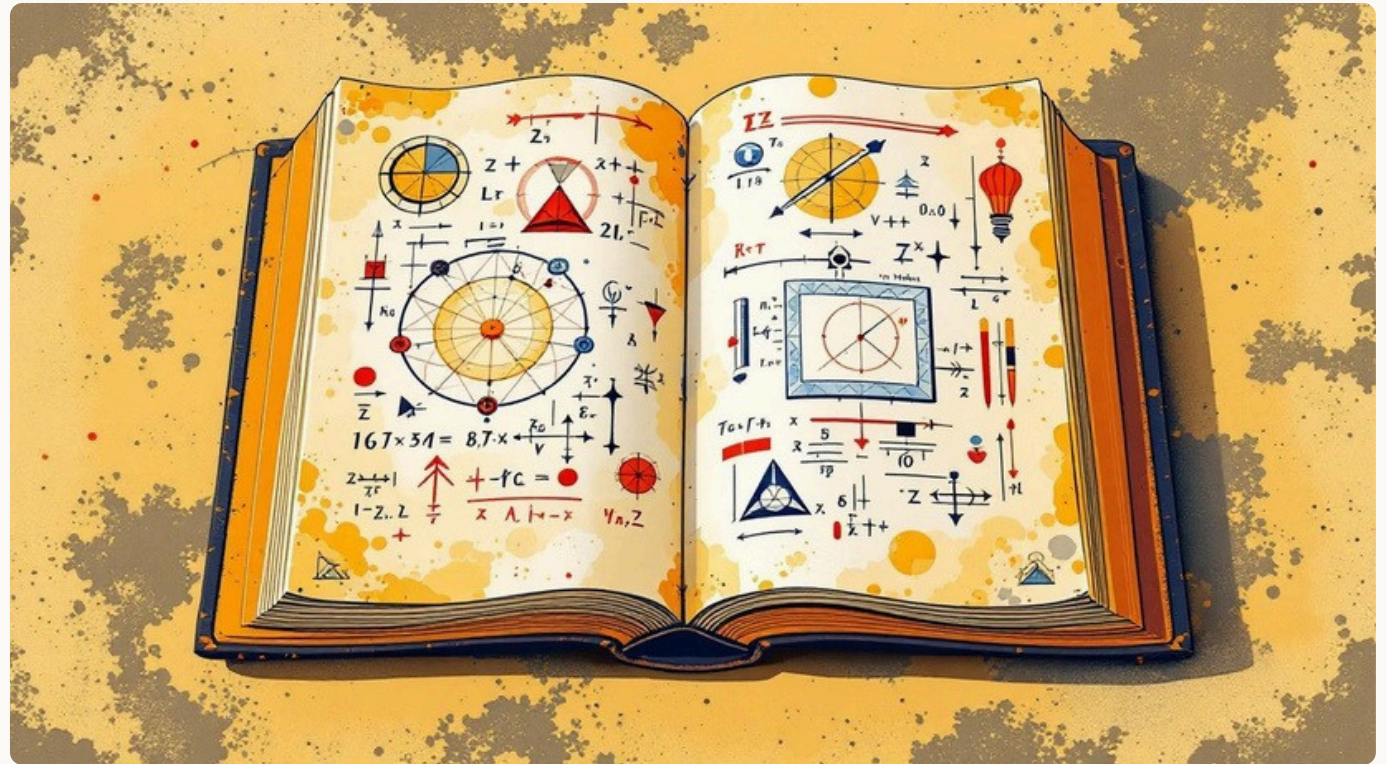
Historical Context

Classic Problem of Antiquity

One of the three famous geometric problems from ancient mathematics, alongside doubling the cube and trisecting an angle.

Baudhayana Sulba Sutra

Compiled between 800–500 B.C.E., this Vedic text provides geometric solutions for altar construction. Section 1.59 addresses circle-to-square transformation.



The Sanskrit Sutra

म लंचतुर ं िचक ष म ो भागा ृा भागमेको न ंशधा िवभ ा िावं शत भागानुरेभाग च षमम भागोनम्प्

01

Ma
ṇḍalam caturasraṇ cikīrṣan

If you wish to turn a circle into a square

02

Viṣkambham aṣṭau bhāgān kṛtvā
Divide the diameter into 8 equal parts

03

Bhāgam ekona-tri
ṇśadhā vibhajya

Take one part and divide it further into 29 parts

04

Aṣṭāviṇśati bhāgān uddharet
Reduce by 28 of those 29 sub-parts

Construction Method



Step 1

Divide circle's diameter (d) into 8 equal parts



Step 2

Take one part and subdivide into 29 equal portions



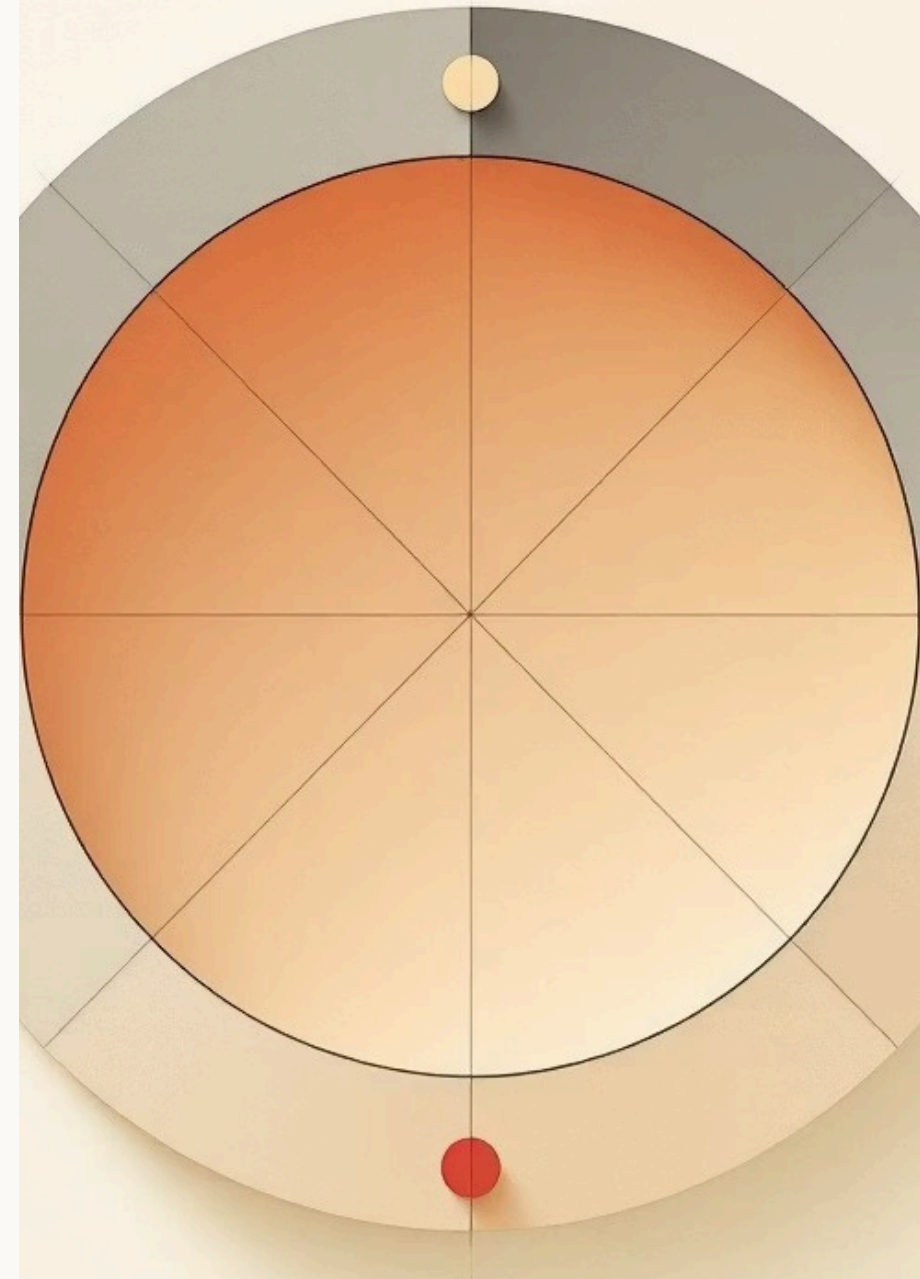
Step 3

Reduce diameter by $\frac{28}{29}$ of that part



Step 4

Apply correction: add one-sixth minus one-eighth of remaining part



Mathematical Formula

Scaling Factor Expression

This side of the square (a) relates to diameter (d) through an alternating series:

$$a = d \left(1 - \frac{1}{8} + \frac{1}{8 \times 29} - \frac{1}{8 \times 29 \times 6} + \frac{1}{8 \times 29 \times 6 \times 8} \right)$$

This yields an approximate value of $\pi \approx 3.088$, sufficient for Vedic brick construction.



Python Implementation

```
def solve_baudhayana_squaring_circle(diameter):  
    """Calculates square side with equal circle area  
    using Baudhayana Sulba Sutra formula.  
    """  
  
    # Define fraction series terms  
    term_1 = 1.0  
    term_2 = 1 / 8  
    term_3 = 1 / (8 * 29)  
    term_4 = 1 / (8 * 29 * 6)  
    term_5 = 1 / (8 * 29 * 6 * 8)  
  
    # Calculate scaling factor  
    factor = term_1 - term_2 + term_3 - term_4 + term_5  
  
    # Calculate square side  
    side_a = diameter * factor  
    return side_a
```

Calculation Steps



Base Unit
1.000000



Subtract $1/8$
-0.125000



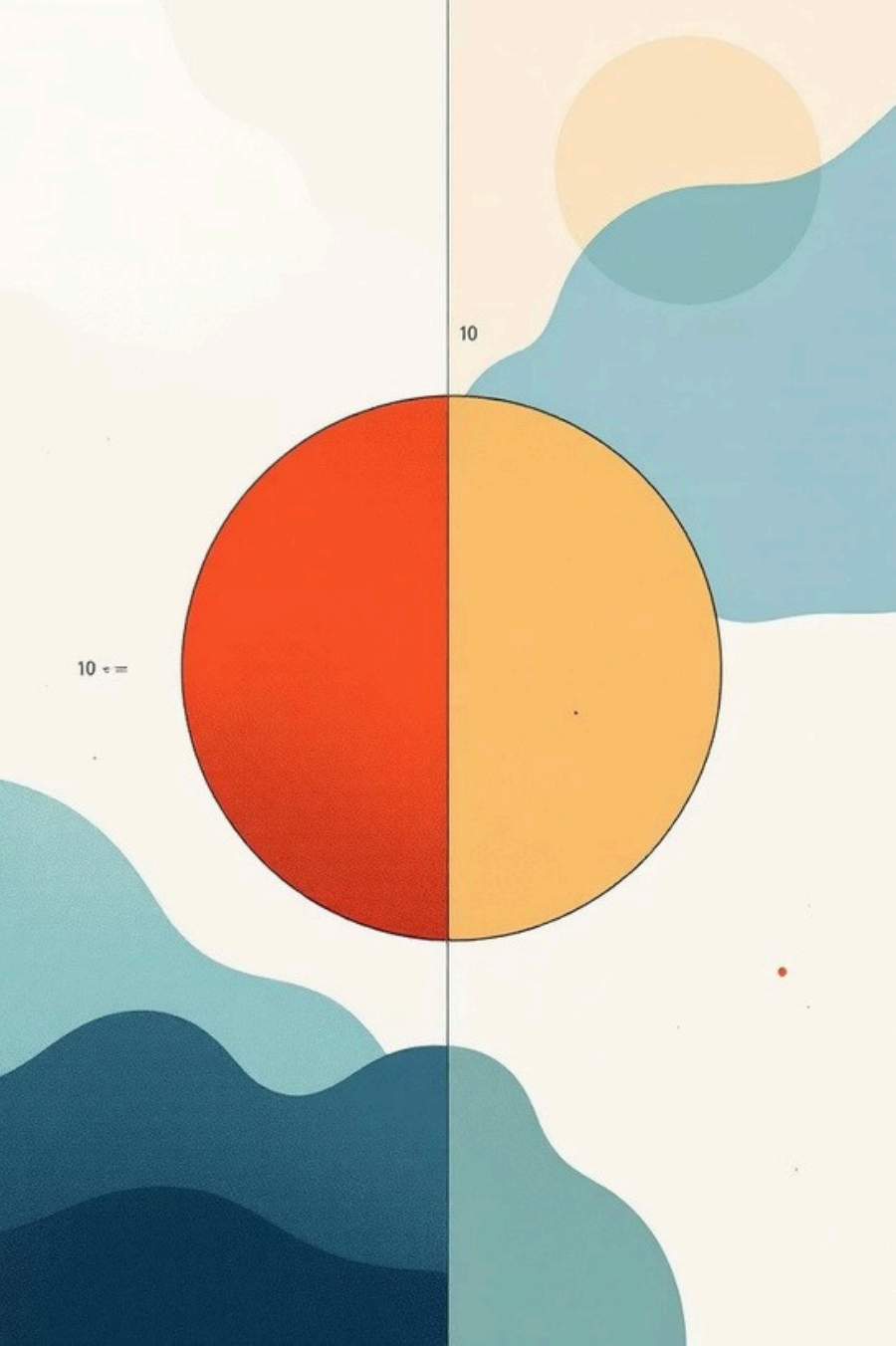
Add $1/(8 \times 29)$
+0.004310



Subtract $1/(8 \times 29 \times 6)$
-0.000718



Add $1/(8 \times 29 \times 6 \times 8)$
+0.000090



Accuracy Analysis

3.0888 3.14... 1.7%

From sutra calculation

Actual transcendental
value

Adequate for
construction

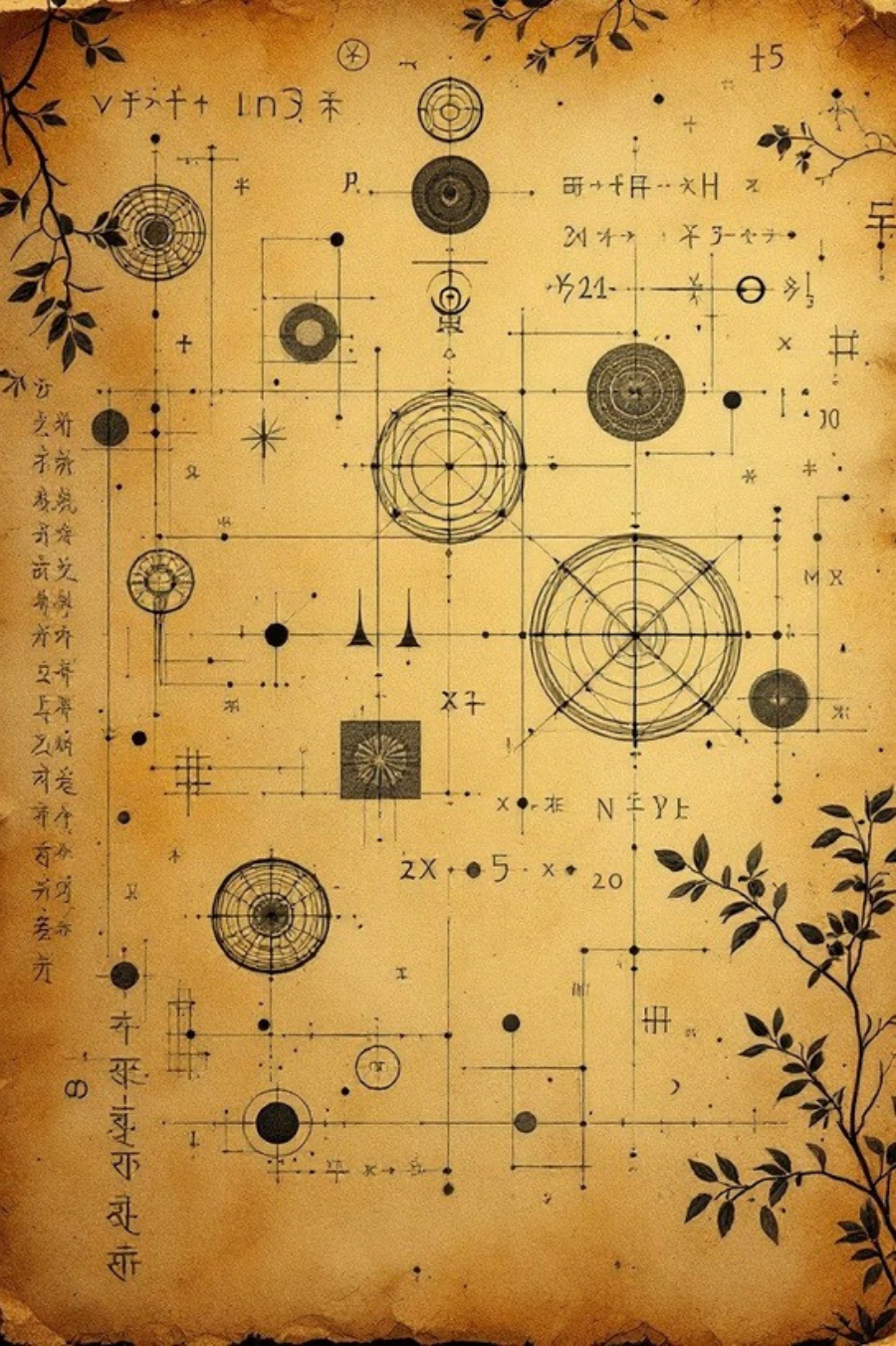
Verification Results

- Diameter: 100 units
- Radius: 50 units
- Modern area: 7853.98 units²

Square Properties

- Side length: 88.776 units
- Area: 7881.18 units²
- Difference: 27.20 units²

- **Engineering Context:** This approximation ($\pi \approx 3.088$) proved sufficiently accurate for Vedic altar construction using fired bricks, where material tolerances exceeded mathematical error.



Conclusion

Historical Significance

Demonstrates sophisticated geometric knowledge in Vedic India centuries before Greek mathematics

Computational Insight

Early example of rational approximation through fraction series expansion

Practical Mathematics

Shows how theoretical geometry served religious and architectural needs